



Society of Petroleum Engineers

SPE-229199-MS

From Spreadsheets to Continuous Monitoring: Water Cut Estimation Using Machine Learning

Khalid Al-Dohaiem, Bibi Hussain Akbar, and Bogi Haryo Nugroho, Innovation & Digitalization Team, Kuwait Oil Company, Kuwait City, Kuwait; Brahmantoyo Aji Sumarto, FD-NFD North Kuwait, Kuwait Oil Company, Kuwait City, Kuwait; Joel Titus, SLB Kuwait, Kuwait City, Kuwait

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229199-MS](https://doi.org/10.2118/229199-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents the transformation of water cut monitoring in Kuwait Oil Company's NFD (BA)-NK field from manual, spreadsheet-based workflows to a real-time, machine learning-enabled system. Traditional practices relied on periodic PGOR tests and manual analysis, which delayed water breakthrough detection and limited proactive reservoir management. As the number of producing wells increased and operational complexity grew, the need for a scalable, intelligent solution became critical. The objective was to automate water cut estimation, deliver faster and more reliable insights, and enable engineers to make timely, data-driven decisions aligned with KOC's digital transformation strategy.

The solution was structured using the OSEMN data science framework—Obtain, Scrub, Explore, Model, and iNterpret. ESP sensor readings and PGOR test data were integrated through automated pipelines connected to the corporate database. Data quality was addressed through routines that handled missing values, normalized variables, and removed outliers using Z-score thresholds. K-Means clustering grouped wells with similar behavior, improving model accuracy and reducing prediction noise. Predictive models were developed using optimized linear regression techniques and automated through Python workflows in Dataiku. Real-time dashboards and alert notifications provided engineers with actionable insights and early warnings of abnormal water cut trends. The system was designed to adapt as new data became available, ensuring long-term flexibility.

The results were substantial. Analysis time dropped from months to hours, predictive accuracy improved with median R^2 values near 0.7, and early detection of water breakthroughs helped prevent significant production losses. Beyond technical performance, the solution streamlined daily workflows, reduced manual effort, and empowered engineers with timely, actionable intelligence. This work demonstrates the value of advanced analytics and automation in modernizing water cut monitoring for mature fields, setting a benchmark for scalable digital oilfield initiatives and proactive production optimization.

Introduction

Accurate and timely water cut (WC) estimation is fundamental for operational excellence and maximizing financial returns in oil production, particularly in mature fields like Kuwait Oil Company's (KOC) NFD (BA) North Kuwait (NFD(BA)-NK) field. The increasing number of producing wells, higher test frequency, and growing operational complexity have historically made water cut surveillance challenging, leading to significant production loss risks from delayed detection.

NFD (BA) - NK was historically dependent upon spreadsheet-based analysis and manual Production Gas Oil Ratio (PGOR) workflows. This was a slow, error-prone, and reactive process that would frequently take weeks per analysis cycle on more than 150 wells. Those kinds of delays prevented proactive decisions and timely intervention, which immediately impacted economies through continuous erosion of possible revenues. Figure 1 illustrates comparison of well performance monitoring without (left) and with (right) water cut prediction, highlighting improved visibility of WC trends alongside pressure parameters.

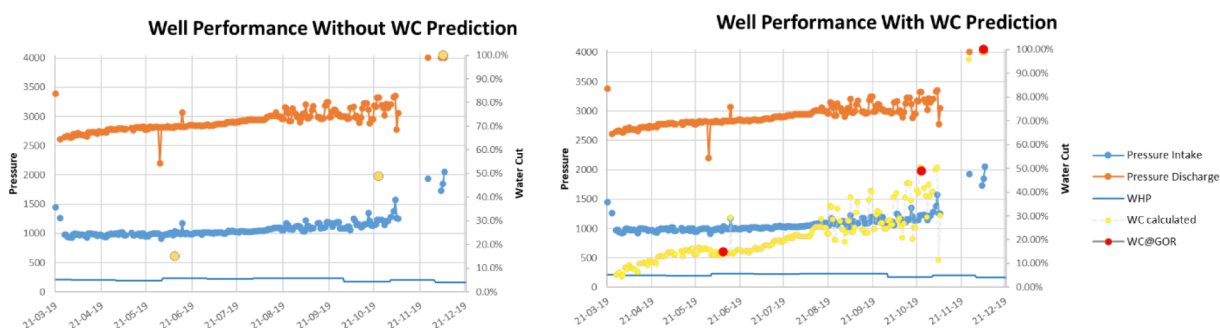


Figure 1—Enhanced Well Performance Monitoring with Water Cut Prediction

The oil and gas sector is now increasingly turning to digital transformation to manage these issues. In this paper, we show that even simple machine learning (ML) approaches can be highly successful when embedded within a comprehensive data science pipeline. The real breakthrough is not just in the ML algorithms, per se, but in the higher-order, structured data science pipeline, which is the Obtain, Scrub, Explore, Model, iNterpret (OSEMN) process, that facilitates their successful application. This project theoretically positions machine learning as interpretable linear regression, constructed within an otherwise robust, operational data science pipeline, highlighting the translation from spreadsheet-centric analysis to real-time water cut measurement.

This paper describes a data science-driven transformation in which real-time, scalable machine learning was integrated seamlessly within a solution for water cut estimation. Aiming for tangible effect and scalability, this strategic digital transformation places KOC in a pragmatic position of digital leadership within the Exploration & Production (E&P) industry.

Field Context and Project Drivers

This idea is inspired by the operational reality of the NFD North Kuwait Field, which has in excess of 150 active wells. At this size, conventional manual approaches, taking months per cycle of analysis, become unviable and unsustainable. The number of wells, combined with time-consuming traditional workflow, increases significantly the difficulty of sustaining optimal production and exposes shortcomings of reactive surveillance.

The fundamental business need behind this project was the need for quicker decision-making, early warning, and scalable insights to efficiently manage production risks and optimize well performance. The financial implications of late water cut detection, which can result in huge production losses, highlighted the extreme significance of this transformation. The total risk caused by cumbersome, manual processes

increases exponentially with the number of wells, rendering the solution inevitable for today's operational size.

This effort directly corresponds to KOC's overall digital transformation program, focusing on real-time, scalable, and trusted decision-making. This dedication to proactive risk mitigation enables a strategic differentiator, transitioning from reacting with responses to avoiding probable losses before their incidence. This proactive capability, enabled through ongoing monitoring, provides a unique competitive advantage to KOC in maximizing asset value and assuring long-term production sustainability. The transition from manual to automated processes, along with the NFD wells' spatial arrangement, is well represented in terms of Figure 2 and Figure 3.

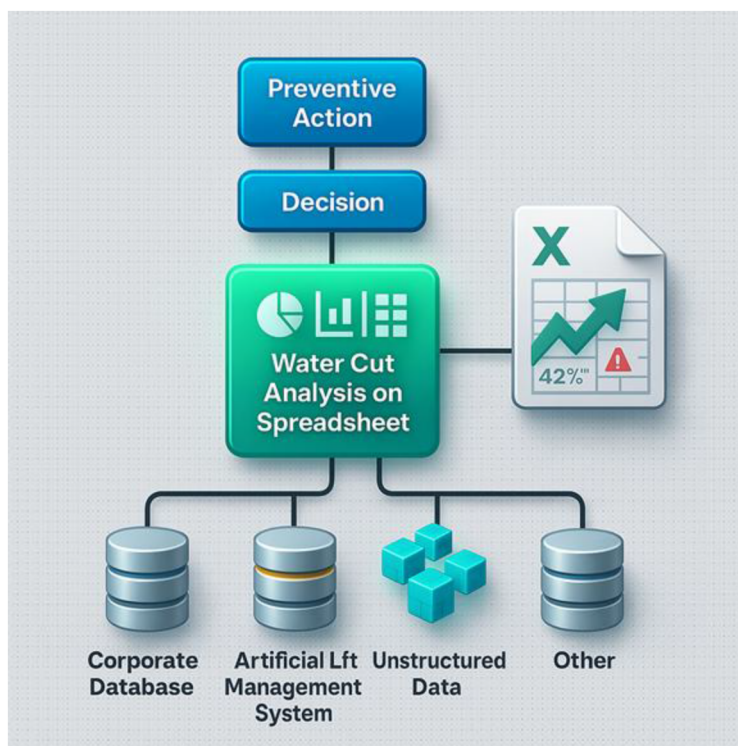


Figure 2—Conventional Water Cut Analysis Workflow.

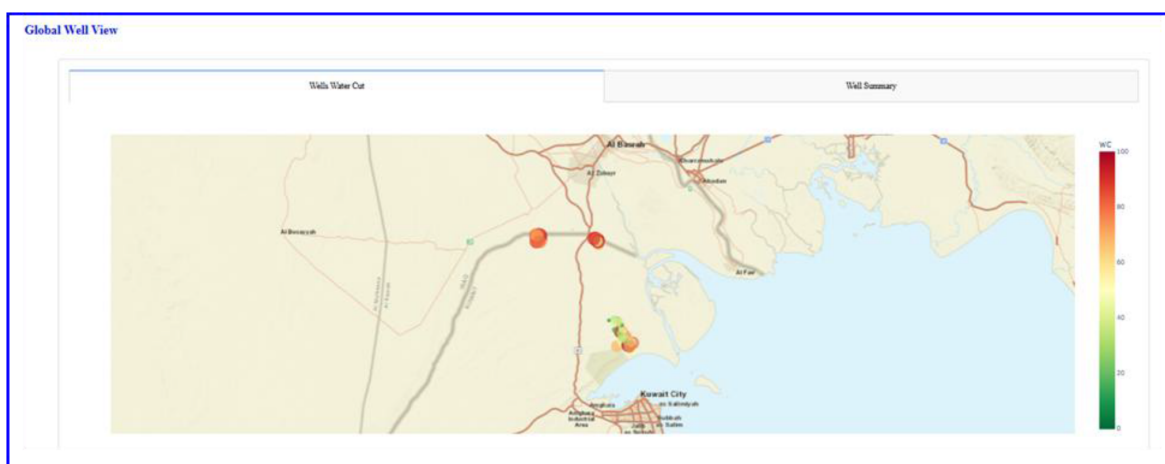


Figure 3—KOC Water Cut Well Data Map

The project also involved a thorough review and enhancement of data quality for water cut estimation across 150 wells in NFD North Kuwait. The complete list of these wells is detailed in [Table 1](#).

Table 1—List of Wells for Data Quality Control and Improvement

Wells	ESP Readings	ESP Depth	PGOR Test	PGOR Salinity	PGOR API	Wells	ESP Readings	ESP Depth	PGOR Test	PGOR Salinity	PGOR API	Wells	ESP Readings	ESP Depth	PGOR Test	PGOR Salinity	PGOR API
14ST2						148H						.55HST1					
18						149H						.56HST1					
19ST1						150H						.57					
20						152H						.60					
23ST1						153H						.63					
25						154H						.68HST1					
28						155H						.69HST1					
29						156H						.70HST1					
30						157H						.72HST1					
33						158H						.73HST1					
34						160H						105					
36						162H						107					
37						163H						.18					
38						164H						.27					
39						165H						.28ST1					
42						167H						.30					
43						169						.31					
47						172						.33ST1					
48						174						.34					
49						179						.35					
50ST1						181H						.43					
51						182H						.44					
55						183H						.46					
58						189H						.47					
62						192H						.48					
63						194H						.52					
65						195H						.54					
67ST1						196H						.56					
69						.05H						.59					
70						.10H						.60					
71						.12HST1						.61					
72						.13H						.62					
73						.14HST1						.63ST1					
74						.15						.66					
75						.18						.67H					
76						.21						.68H					
81						.26HST1						.69ST1					
82ST1						.29HST1						.70HST3					
83						.31HST1						.71					
84ST1						.33HST1						.79					
86ST1						.34HST1						.81					
89ST2						.37HST1						.83H					
02						.39HST1						.86					
06						.40HST1						.88					
24						.41H						.89					
26						.42H						.92ST1					
38						.43HST1						.93ST1					
39H						.46						.96					
43						.53HST1						.98					
45						.54HST1						.01					

Structured Workflow: Data Science with OSEMN

The successful implementation of this water cut estimation solution is a testament to a rigorously engineered, structured intelligence approach, moving beyond ad-hoc machine learning applications. This project systematically utilized the Obtain, Scrub, Explore, Model, iNterpret (OSEMN) framework, originally developed by Mason and Wiggins in 2010. This framework provided a comprehensive and systematic pathway for transforming raw data into actionable operational insights. The structured flow of this data

science process, from initial data acquisition to final interpretation, is a critical component of its success and is conceptually depicted through Figure 4.



Figure 4—Turning Conventional Workflow to Data Science Driven Decision Making

Obtain: Data Acquisition and Integration

The foundational step involved meticulous acquisition and integration of diverse datasets. Primary data sources included Electric Submersible Pump (ESP) sensor readings and Production Gas Oil Ratio (PGOR) test results, extracted directly from KOC's corporate Oracle database, incorporating intelligent data pulling and secure access protocols.

Supplementary information, such as formation tops, fluid properties (salinity, API gravity), and spatial data (field boundaries, well locations), was recognized as crucial for enhancing analytical results and model accuracy. While not all supporting data were fully utilized in this phase, their potential for future enhancements was acknowledged.

The reliability of any data-driven model is tied to the quality and completeness of its input. Figure 5, a comprehensive data availability metric chart highlights the status of various data types for the 150 assessed wells.

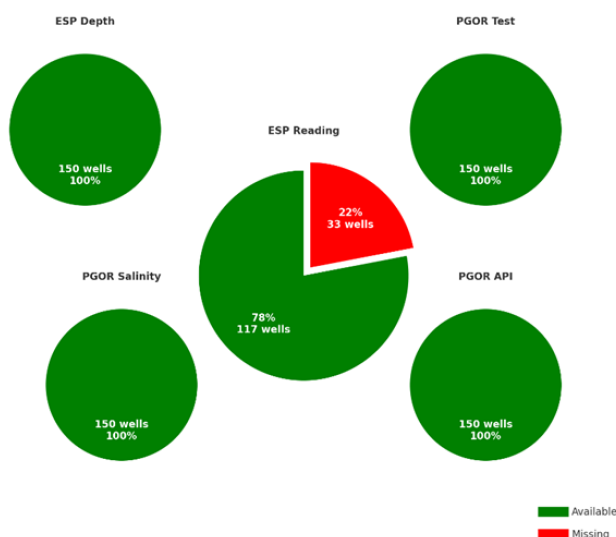


Figure 5—Data Availability Metric Chart

This metric revealed a notable gap: 22% of ESP Reading entries (33 wells) were missing, while other parameters—ESP Depth, PGOR Test, PGOR Salinity, and PGOR API—demonstrated 100% completeness. This underscores that the Obtain phase extends beyond merely acquiring data; it critically involves ensuring high-quality, complete data for consistent model performance. Challenges like sensor data gaps necessitate fallback data handling strategies and highlight the importance of robust data governance for KOC to truly operationalize data science at scale.

Scrub: Data Quality and Preprocessing

This stage required extensive data preprocessing and cleaning to be accurate and reliable. This included automated procedures to account for missing values, standardize variables, and detect outliers via Z-score filtering. It followed the ‘garbage in, garbage out’ mantra, wherein inherent raw data flaws were not transmitted to become unreliable analytical results. This stage also involved frequent ‘human-in-the-loop’ feedback loops with domain specialists to optimize data representations, like eschewing deceptive visual interpolations for missing information. This human-in-the-loop approach was crucial for building trust and operational meaning.

Explore: Exploratory Data Analysis and Feature Engineering

The Explore stage focused on uncovering data patterns and relationships. This involved comprehensive Exploratory Data Analysis (EDA) and the application of K-Means dynamic clustering methods to group wells with similar behaviors. These clusters were categorized into distinct water cut regimes (low, medium, and high WC), acknowledging that a single model may not be universally applicable. Engineers played a pivotal role in validating these clusters, ensuring alignment with domain expertise. This collaborative approach was vital for creating features that were both statistically robust and physically meaningful. The resulting operational stratification enabled tailored strategies for each well cluster, optimizing resource allocation and management. The insights from this phase are visually articulated through [Figure 6](#).

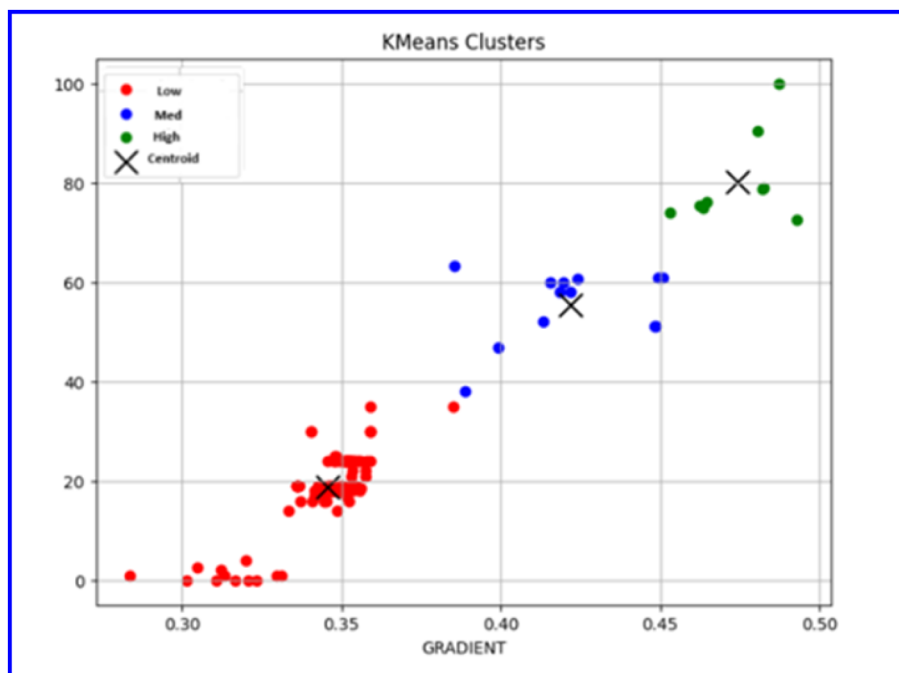


Figure 6—Data Cluster using K-Means Dynamic Clustering for Pattern Understanding and Trend Identification - Well XX-0007.

Model: Linear Regression-Based Machine Learning for Water Cut Estimation

The core of the solution involved developing and training simple yet scalable linear regression models within the Dataiku platform. These models were tailored for each well cluster, prioritizing transparency and interpretability to foster trust among engineers. The entire modeling process was rigorously automated using Python-based workflows, enabling daily runs and significantly boosting value through repeatability and scalability. Robust automated function optimization was a cornerstone, involving hyperparameter tuning and the enforcement of predefined conditions to ensure alignment with petroleum engineering principles. This approach was fundamental to overcoming the ‘trust gap’ and empowering engineers to understand and interact with the models.

Interpret: Actionable Insights and Real-Time Alerting

This stage involved conveying model outputs as clear, actionable information. This required setting up accurate alert thresholds and delivering intuitive dashboards with water cut trendlines and anomaly flags. A key success was shortening the data-to-action cycle from months down to minutes, giving engineers timely, educated decisions. Automated warning and real-time dashboards decreased engineers’ need for manual data extraction considerably, changing their role from reactive data extractors to proactive asset managers. The system configuration for such alerts is visualized through Figure 7, and the system for such alerts is visualized through Figure 8.

Mail (SMTP) channel

ID

WC1

Host

Port

25

Use SSL

☐

Use TLS

☒

Extra properties

+ ADD PROPERTY

Additional properties for the mail session

Login

@kockw.com

Password

Sender email address

@kockw.com

*From" address for mails sent with this channel. When set, this address is used for all mails sent with this channel and cannot be overridden.

Authorized recipients domains

kockw.com

Comma-separated list of authorized domains for "To" addresses. Examples: scme.com, yourcompany.com

Figure 7—Alert Configuration Flow Diagram

Global Well View

Wells Water Cut

Well Summary

Search Title

Export

Filter

ID	WC	Clusters	Low_Z_min	Low_Z_max	Low_Slope	Low_Intercept	Low_R2	Low_acc	High_Z_min	High_Z_max	High_Slope	High_R2	High_acc
AD-0848	100	['High', 'Low']	-0.750000000000000	-0.5	5208.413	-1775.177	0.982	93.047	No enough data				
BH-0157	100	['High', 'Low', 'Med']	-3.0	-0.5	1461.199	-507.811	1	100	No value for optimized				
RQ-01670	99.117	['High', 'Low', 'Med']	-1.200000000000004	-0.15	1578.767	-500.332	0.822	84.432	-3.0	0.6	481.198	-117.373	0.777
BH-08620	97.493	['High', 'Low', 'Med']	-1.300000000000000	-0.05	1018.455	-138.425	1	100	-3.0	0.7	122.517	-20.764	0.304
RQ-01830	94.294	['High', 'Low', 'Med']	No value for optimized						-3.0	1.5	477.887	-129.476	0.792
BH-0072	93.356	['High', 'Low', 'Med']	No enough data						-3.0	1.35	275.814	-68.55	0.405
AD-0033	92.663	['High', 'Low', 'Med']	No value for optimized						-1.450000000000005	0.9	276.281	-66.327	0.47
AD-0081	91.711	['Low', 'Med']	No enough data						-1.500000000000003	0.9	335.813	-71.585	0.41
AD-0052	91.08	['High', 'Low', 'Med']	-1.200000000000004	1.25	645.316	-225.383	0.805	-30.972	-3.0	1.7	466.899	-148.797	0.745
AD-0038	91.416	['High', 'Low', 'Med']	0.540000000000000	1.35	695.581	-250.93	0.445	49.582	-3.0	-1.2	124.697	-0.48	0.736

Figure 8—Detailed Water Cut and Cluster Model Summary

Deployment and System Architecture

Advanced solution was deployed across all 150 wells within the NFD (BA) - NK area. System design, from automated data pipelines through Dataiku's recipe-based methodology, monitors the overall process from ingestion to modeling and incremental updates. The fault-tolerant design yielded consistency, precision, and velocity, with a high level of manual intervention being minimized. Collaborative beta testing with field engineers helped refine and instill confidence, both to propel the system towards operational needs of real-world scenarios.

The system is scalable from within, with inherent capability to self-adapt new wells upon expansion. Strong fail-safes, such as fallback rules for sensor failure, were also deployed to give faultless predictions, enhancing reliability of system in an ever-changing scenario. Extent of enhancements and system design is tabulated in [Tables 1 and 2](#), and graphically represented in [Figure 9](#) and [Figure 10](#). This represents a transformative shift from conventional, spreadsheet water cut estimation methods to a continuous, preventive monitoring approach, enabling proactive decision-making and enhanced reservoir management.

Table 1—Solution Key Functionalities

Feature	Pilot Version (Input)	Enhanced Version (Input)
User Defined Filter	Installation 1, 2 relate to ESP installation - simplify to start and end date. W/O, new ESP installation based on event/data. Data selection, pre-processing and data analysis. User defined water cut: PGOR and samples.	Installation: control based on date/event, limitation: pump installation date: relate to f function. ESP installation is locked after update. Automated Statistical Based Optimization. Pilot Version + Enhance with Full OSEMNI Implementation Framework (Statistic Sum, Transform, Outlier and Anomaly detection, Z-score, K-Means dynamic clustering, Processing optimization) + Automated, Semi.
Model Build	Build model with min 2 PGOR data. If there is only one, use water cut from test.	User Preference Model and Manual Model Optimization.
Formula & Parameters	Tubing size affecting f, liquid rate is mandatory in data. Tubing size and condition related to the formula.	Auto update model based on start - end date and data range (simple exploratory data analysis).

Table 2—User Interface Components of Enhancement System

Category	Sub-Category	Pilot Version Component	Enhanced Version Component
Filters, model, and chart parameters	Data Filter	Well selection (Yes), Date Interval selection (Yes)	Well selection (Yes), Date Interval selection (Yes)
	Water Cut Model Parameters	Sample selection (Yes), ESP depth range (Yes), Tubing size (Yes), Tubing condition (Yes)	Sample selection (Yes), ESP depth range (Yes), Tubing size (Yes), Tubing condition (Yes)
	Chart Parameters	Y-Axis display range (Yes), X-Axis display range (Yes)	Y-Axis display range (Yes), X-Axis display range (Yes)
	Model Fine-Tune	WC-Gradient slide bar	Explore (Statistic-based user preference, Lasso (Manual user preference model))
Visualization	Charts	Water Cut Prediction (by Time) (Yes), Gradient Plot (Yes)	Water Cut Prediction (by Time) (Yes), Gradient Plot (Yes)
	Tabular	Water Cut Summary (by Well) (Yes, by Well + Multiple), PGOR test (Yes, by Well + Multiple), ESP readings (Yes, by Well + Multiple), Last water cut by well (Yes, by Well + Multiple)	Water Cut Summary (by Well) (Yes, by Well + Multiple), PGOR test (Yes, by Well + Multiple), ESP readings (Yes, by Well + Multiple), Last water cut by well (Yes, by Well + Multiple)
	Chat, Plot Enhancement	No	Box plot, Performance metrics
Map	Water Cut Status Map	No	Yes
Optimization	Model and Processing	No	Statistic and Data resolution

Category	Sub-Category	Pilot Version Component	Enhanced Version Component
Email Alert	Email notification + attachment for details	No	Yes



Figure 9—System Architecture Diagram

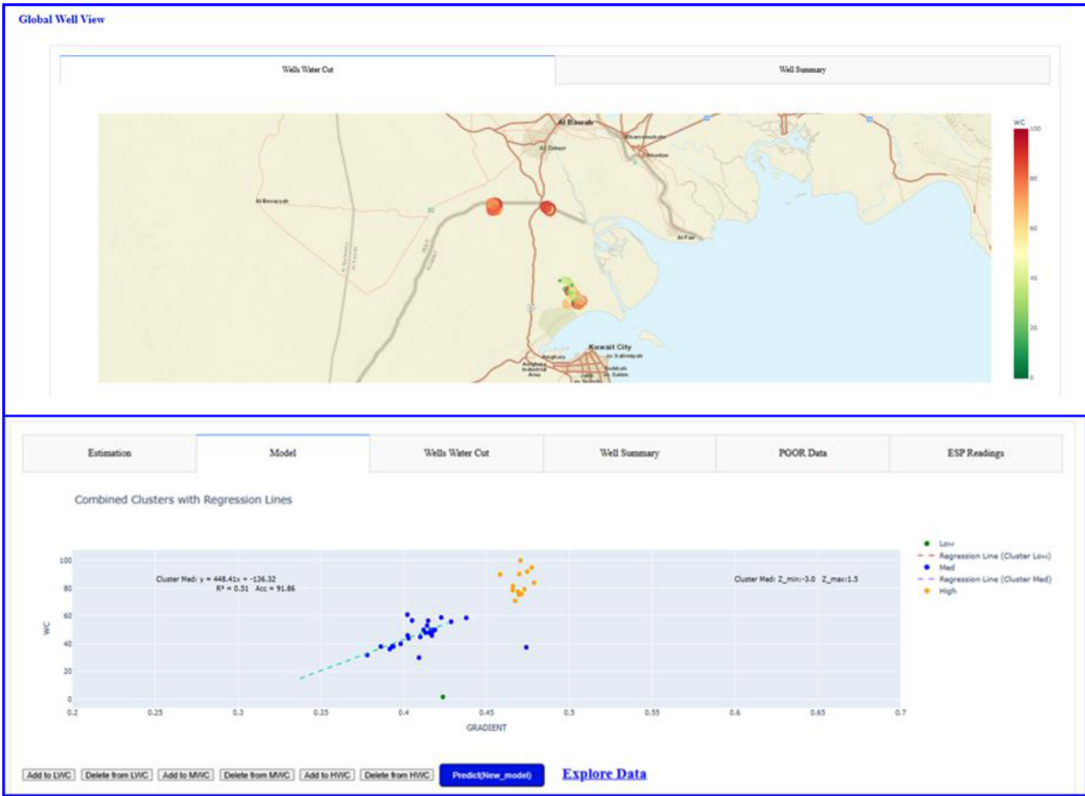


Figure 10—Enhanced Water Cut Monitoring and Predictive Analytics System - Fascinating View

Results and Field Validation

The project's value is measured through extensive efficiency, accuracy, and economic value enhancements. Full-field implementation achieved a tremendous analysis time reduction from more than two months to fewer than two hours for 150 wells, an unprecedented 5,400-fold increase. The predictive models yielded stable and consistent outcomes, reaching an average R-squared (R^2) value of around 0.7. Performance analysis per water cut regime indicated high correlation for low and high water cut wells, and the medium water cut regime indicated opportunities for further optimization. This offers strategic guidance for future optimizations. This overall across-beta and across-full-field consistency in accuracy and processing time emphasizes reliability and scalability of the solution.

Live monitoring and early warning allowed for quicker interventions, and 1.29 million barrels of oil production loss were prevented, worth about USD 77.2 million in value retention. Automated availability of water cut forecasts and visual well summary information helped engineers significantly, reducing their need for manual data collection and changing their function from passive data consumers to proactive asset managers. A graphical representation of the economic value impact is illustrated in [Figure 11](#), while the effectiveness of the alert system in detecting abnormal water cut trends is shown in [Figure 12](#).

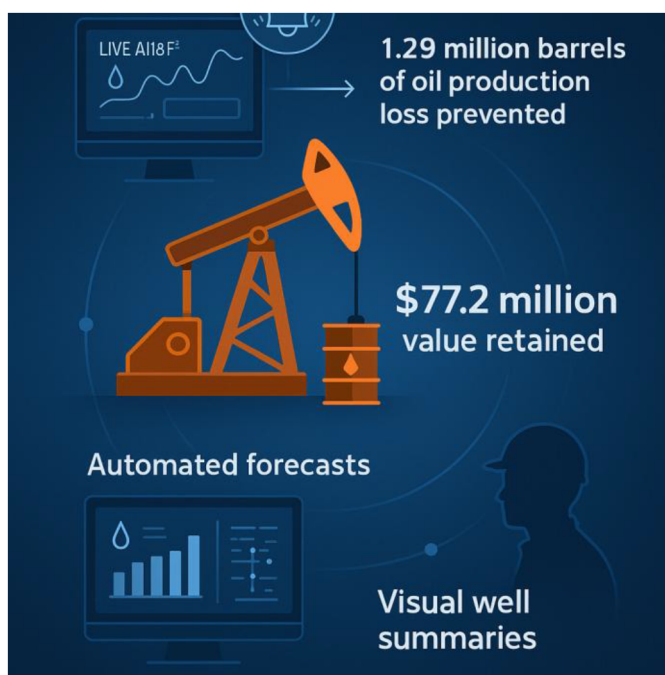


Figure 11—Economic Impact Infographic

Water Cut Monitoring Report(Monthly_level)

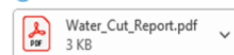


Krupakar, Joel Titus

To: Nugroho, Bogi Haryo; Sumarto, Brahantiyo Aji; Hartono, Sugiarto; Krupakar, Joel Titus

Sun 3/2/2025 1:24 PM

🔗 If there are problems with how this message is displayed, click here to view it in a web browser.



Dear Team,

Please be informed that an estimated increase in watercut levels has been observed for the following wells :

UBHI	CHANGE IN WC PREDICTION IN LAST ONE MONTH (%)
XX-005YY	4.30
XX-009YY	10.00
XX-013YY	11.19
XX-015YY	11.01

We recommend closely monitoring these wells and taking necessary actions to mitigate any potential impact on production.

For details please find attached file. If you have any questions or require further assistance, please do not hesitate to contact us..

Best Regards,

Watercut Integrated Monitoring System

Figure 12—Water Cut Prediction Email Alert

Challenges and Lessons Learned

A transformative digital solution's implementation always poses its set of challenges. A major challenge was how to address sensor data gaps, considering 22% of ESP reading entries were absent. This was alleviated through engineered fallback logic and real-time data quality verification.

A key non-technical challenge was engineers trust gap. This was breached through model transparency being a high priority, setting up feedback loops, and granting engineers tuneability and inspection of the models. The project also achieved a model complexity trade-off, opting for much simpler, interpretable regression models over more complex methods to boost user adoption and ease of integration.

These lessons highlight the critical importance of integrating human-lefted thinking with technological advancements to drive meaningful progress. By balancing innovation with usability, organizations can deliver solutions that truly address stakeholder needs. Table 3 provides a structured summary of the major challenges faced, the strategic actions taken, and the measurable results achieved.

Table 3—The Summary of Key Challenges, Actions, Outcomes

Challenge	Action	Outcome
Sensor data gaps (e.g., 22% missing ESP Readings)	Implemented engineered fallback logic; Continuous data quality validation	Resolved data gaps, maintained system functionality
Trust gap / Engineer skepticism of automated models	Continuous validation; Feedback loops with field teams; Enabled engineer tuning/overriding of models	Increased engineer adoption; Built confidence in system reliability
Model complexity vs. adoption trade-off	Prioritized simpler, interpretable regression models	Enhanced explainability and user acceptance; Successful integration into workflows
Need for continuous improvement	Established ongoing feedback loop with field teams; Iterative system refinement	System remains relevant and effective; Fostered culture of continuous improvement

Scalability, Roadmap, and Future Enhancements

The successful implementation and validation of the water cut estimation solution in the NFD (BA) - NK field establishes a robust foundation for future expansion and strategic enhancements. The project's methodology and architecture are designed to be a replicable blueprint, facilitating the modernization of production surveillance across other fields within KOC and potentially beyond. This positions the project as a foundational component of KOC's enterprise-wide digital strategy.

Future development roadmap entails a strategic extension to additional key surveillance parameters, including possible monitoring of Electric Submersible Pump (ESP) performance and Gas-Liquid Ratio (GLR), building upon the developed data science infrastructure and pipeline.

Looking further in the future, the project envisions adding more advanced Artificial Intelligence (AI) and Machine Learning (ML) functionality. This would encompass investigating development of a Generative AI explanation layer for deep operational questions. Of further major enhancements is adding real-time shutoff logic to automated well control. Additionally, there is promise for even deeper integration with digital twin environments or advanced Production Operations dashboards to provide an integrated, real-time, holistic overview of asset performance and predictive functionality. These advanced features point to a forward-looking strategy, where KOC is actively investigating and preparing for next-generation solutions to stay ahead competitively.

Scalability of the project was rigorously evaluated when it was rolled out across 150 wells, with equal emphasis on accuracy as well as processing efficiency. Consistent and repeatable prediction accuracy achieved during beta testing (40 wells) and complete rollout (150 wells) is a strong indication of the solution's scalability with respect to accuracy.

Time taken to process 150 wells was optimized to be extremely efficient, computing all of the dataset within 120 minutes, which turned out to be an average of 1 minute per well. It was equivalent to beta test performance, which indicated scalability since it handled ramped-up workloads efficiently through parallelized computation, being poised for real-time uses. This attests that the solution can easily adapt to large datasets without sacrificing performance, both in accuracy and time of processing.

Conclusion

The water cut estimation project in NFD (BA) - NK has set a new benchmark for operational excellence in the E&P sector. By leveraging a foundational linear regression machine learning model within the disciplined OSEM data science framework, the team delivered a solution that is accurate, highly scalable, and fully transparent earning the trust and adoption of engineers across the field. The impact is both measurable and transformative, reflected in key outcomes:

- Analysis speed accelerated by 5,400x
- Forecasts consistently precise
- High adoption across engineering teams
- Potential USD 77.2 million in lost production prevented

This achievement proves a critical truth that scalable AI begins with strong data science. More than technical success, this initiative establishes a replicable blueprint for the E&P industry, demonstrating that data-driven innovation unlocks massive value, enables real-time monitoring, and positions organizations at the forefront of digital transformation.

Acknowledgements

The successful completion of this paper is a testament to the collaborative efforts of multiple teams and individuals. Special gratitude is extended to the KOC Innovation & Technology Management, Innovation & Digitalization Team for their vision and support, and to the NFD (BA) North Kuwait for their invaluable contributions and feedback. We also acknowledge the support from SLB partners.

References

1. Akbar, B. H., Al-Aradi, H. J., Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., & Achmad, P. R. *Harnessing Data Science for Enhanced Water Cut Analysis in NK-NFD*. Kuwait University Scientific Poster Day, Kuwait, February, 2025.
2. Nugroho, B. H., Akbar, B. H., Sumarto, B. A., Al-Aradi, H. J., Joenaedy, M. A., & Achmad, P. R. *Harnessing Data Science for Produced Water Estimation: Unleashing the Power of Data-driven Techniques*. SPE-219016-MS, SPE Water Lifecycle Management Conference & Exhibition, Abu Dhabi, March, 2024.
3. Akbar, B. H., Al-Aradi, H. J., Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., & Achmad, P. R. (2024, December). *Accelerating Insights Automated Water Cut Estimation for Enhancing Production Optimization*. KPC and Google Cloud event, Kuwait.
4. Al-Aradi, H. J., Nugroho, B. H., Akbar, B. H., Sumarto, B. A., Joenaedy, M. A., & Achmad, P. R. *Data Science Driven Decision Making Insight into Water Ratio Production Level*. Middle East Water Week Conference and Exhibition, Al Khobar-KSA., December, 2023.
5. Sumarto, B. A., Nugroho, B. H., Akbar, B. H., Joenaedy, M. A., Al-Aradi, H. J., & Achmad, P. R. *Water Cut Estimation Using ESP Sensor Reading and Well Test Data, A Novel Approach*. SPE-211407-MS, ADIPEC, Abu Dhabi. November, 2022.
6. Akbar, B. H., Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., Al-Aradi, H. J., & Achmad, P. R. *Data Science For Water Cut Analysis System Utilizing ESP Sensor Data Prototype, Pilot and Its Potential Impact*. SPE-211406-MS, ADIPEC, Abu Dhabi, November, 2022
7. Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., Al-Aradi, H. J., & Achmad, P. R. *Data Science for Water Cut Analysis System Utilizing ESP Sensor Data, Conceptual Model, and Its Proposed Solution*. SPE-207624-MS, ADIPEC, November, 2022.
8. Nugroho, B. H., Sumarto, B. A., Akbar, B. H., Titus, J. *Data Science for Water Cut Estimation in NFD (BA) - NK Phase 4 – Field Deployment Project Report*. KOC, Kuwait, March, 2025.
9. Nugroho, B. H., & Achmad, P. R. *Data Science for Water Cut Estimation in NFD (BA) - NK Phase 2 & 3 - Data Quality Control, Prototyping & Pilot Project Report*. KOC, Kuwait, March, 2021.
10. Nugroho, B. H., & Achmad, P. R. *Data science for Water Cut Estimation in NFD (BA) - NK - Phase 1 Assessment Project Report*. KOC, Kuwait, January, 2021.
11. Sumarto, B. A. *ESP General Operation and Technical Procedures*. *Personal proprietary and limited internal disclosure*, Kuwait, 2019.
12. Eric, J., *Data Science Life Cycle: CRISP-DM and OSEMN frameworks*, Website, *Data Science Life Cycle: CRISP-DM and OSEMN frameworks*.
13. Data Science PM, *OSEMN Data Science Life Cycle*, Website, OSEMN Data Science Life Cycle Data Science PM, March, 2024.
14. Damlapinar, M. *Analytics of Life: Making Sense of Data Analytics, Machine Learning and the Artificial Intelligence*. NLITX, New York, 2019.
15. Hastie, T., Tibshirani, R., & Friedman, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer, New York, 2009.

16. Rahman, R. *From Information to Impact: How Data Drives Decision-Making*. DEV Community, October, 2024.
17. Liashchynskiy, P., & Liashchynskiy, P. *Grid Search, Random Search, Genetic Algorithm: A Big Comparison for NAS*, Cornell University, New York, 2019.
18. Bergstra, J., & Bengio, Y. Random Search for Hyper-Parameter Optimization, *Journal of Machine Learning Research* **13**, p: 281-305, 2012.
19. Mujahid Tabassum, Kuruvilla Mathew., A Genetic Algorithm Analysis towards Optimization Solutions, *International Journal of Digital Information and Wireless Communications (IJDIWC)*, **4** (1), p: 124-142, January, 2024.
20. Institute of Data. *How Businesses Can Best Use Data Science to Support Operations*, April, 2024.
21. *The data science venn diagram*: <http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram>, March, 2013.
22. *Microsoft Excel Limitation*: <https://www.ibm.com/support/knowledgecenter/en/SSD29G2.0.0/com.ibm.swg.ba.cognos.tmlinst.2.0.0.doc/cexcellimits.html>.